
KTU
NOTES

denoting loss of synchronism or falling

acteristics can be analyzed in terms of
speed of the system.

ed in terms of rotor angle, δ

o three basic types-steady state,
transient.

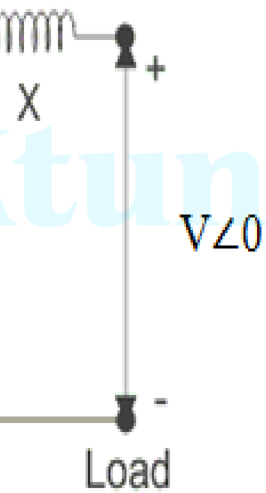
ssumed to be applied at a rate
w when compared either with the
uency of oscillations of the major
system

e stability

m flow of power possible through a
ut losing the stability with sudden
anges in the network conditions by
dden large increment of loads.

e two is that dynamic stability is
ble by the action of fast acting
lators which are capable of
e flux at a faster rate than that
he system in falling out of step
steady state stability we assume
ulator acts slowly in order to
erminal voltage to the prescribed

l-by using power flow equations.



$$S = VI^*$$

Neglecting R

$$= |V|\angle 0 \left[\frac{|E|\angle\delta - |V|\angle 0}{|X|\angle 90} \right]^*$$

From this

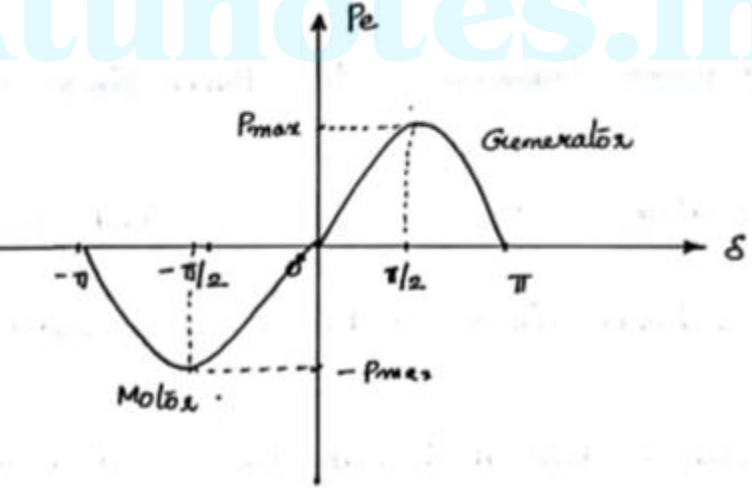
$$P_e = \frac{|E||V|}{|X|} Sin\delta$$

$$If \;\; \delta = 90^\circ$$

$$P_e = P_{\max} = Steady\ State\ Stability\ Limit$$

$$P_{\max} = \frac{|E||V|}{|X|}$$

s generator may lose synchronism.
motor also.



$$= \frac{EV}{X} Cos\delta$$

$$> 90^0$$

is negative and unstable.

ed

stem at a higher level of voltage

e reactance of the system

nes, Double circuit lines, Bundled

s, Series capacitors

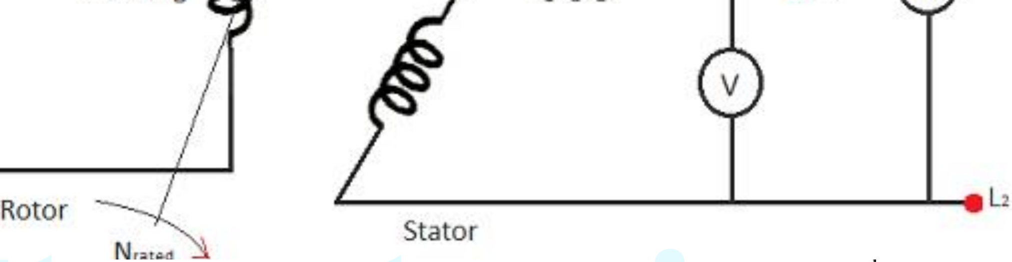
rrence of fault, there may be a
e rotor position w.r.to stator, so
or of the rotor is required.

rator is represented by its
equivalent.

enoted by δ and depends upon the
machine. Larger the loading, larger is the
rque angle δ.

added or removed from the shaft of the
achine, the rotor will decelerate or
pectively with respect to the
rotating stator field and a relative motion
d that the rotor is swinging with respect to
l.

describing the relative motion of the rotor
with respect to the stator field as a
e is known as swing equation.

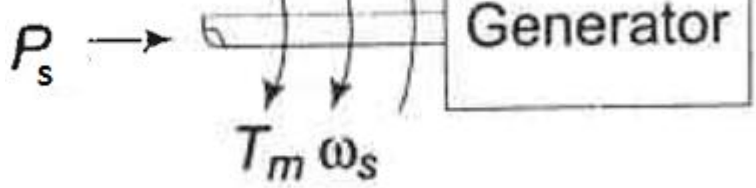


$\nearrow E$ corresponding to rotor axis.

δ

$\rightarrow V \Rightarrow$ corresponding to stator axis.

(synchronously rotating).



leration nor deceleration

occurs, $P_e \downarrow$, *But no corresponding change in* P_s

nd $+ve$

rts accelerated.

Power $P_a = P_s - P_e$

$oment\ of\ Inertia, MJ\ \sec/\ Electrical\ \deg ree$

$rtia, Kgm^2$

$= \frac{d^2\theta}{dt^2}$

uously with respect to time when a sudden change occurs in electrical degrees from the synchronously rotating reference axis.

$$\frac{d\theta}{dt} = \omega_r + \frac{d\delta}{dt}$$

$$\frac{d^2\theta}{dt^2} = \frac{d^2\delta}{dt^2}$$

$$P_a = P_s - P_e = P_s - \frac{EV}{X} Sin\delta$$

Swing Equation

$\mathrm{I}f)$

nd / radian

ond / electrical deg *ree*

ons tan *t is also represented with M*

size of the machine as well as on its type

vary widely with size and has a characteristic

s for each class of machines.

olution

machine system

first swing of the machine.

will be maximum during first swing

table during first swing , it will be stable in subsequent

pplied for multi machine systems.

chances of loosing stability in subsequent swings.

$$P_a = P_s - P_e = P_s - \frac{EV}{X} Sin\delta$$

sides of the equation by 2dδ/dt, and integrating

ne.

$$-P_e)\frac{d\delta}{dt}dt$$

$$\left[M\left(\frac{d\delta}{dt}\right) \right]^2 = 2\int (P_s - P_e)d\delta$$

$$P_e)d\delta$$

$$\frac{d\delta}{dt} = \sqrt{\frac{2}{M}\int (P_s - P_e)d\delta}$$

initial torque angle before any

occurs and at this time $d\delta/dt = 0$.

will stop changing and the

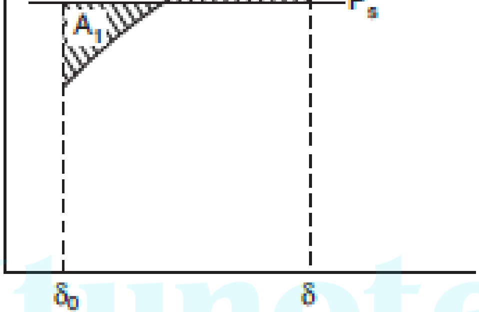
l again be operating at

s speed after a disturbance when

et it happens at an angle δ

$$= 0$$

$$\int_{\delta_0}^{\delta} (P_s - P_e)\,d\delta = 0 = \int_{\delta_0}^{\delta} P_a\,d\delta$$



the curve Pa should be zero which is possible
both accelerating and decelerating powers.

Ps > Pe for positive area A1 and

Pe > Ps for negative area A2 .

lerating Area

decelerating Area

OR

overed during acceleration is equal
red during deceleration”

Stable

Critically Stable

Unstable

and $\omega=\omega s$ or $Pa=0$ and $\omega \neq \omega_s$

$\omega>\omega s$, δ increases.

$\omega<\omega s$, δ decreases.

at the middle of the Transmission
allel transmission network

on a line near to busbar in a
smission network

on the busbar in a parallel
n network

Pe1

Pe2

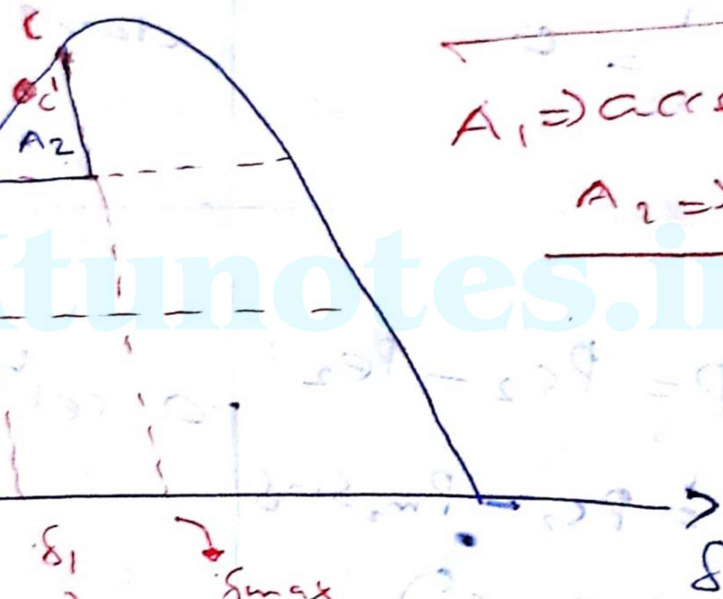
$$\frac{EV}{X_{1eq}} Sin\delta_0 = P_{m_1} Sin\delta_0$$

$$= X_g + X_l$$

$$\frac{EV}{X_{2eq}} Sin\delta = P_{m2} Sin\delta$$

$$= X_g + X_l = X_{1eq}$$

$$P_{m1}$$



$A_1 \Rightarrow$ accel^n region

$A_2 \Rightarrow$ deccl^n region

δ_1

$\omega = \omega_{max}$

(or)

$\omega = \omega_{min}$

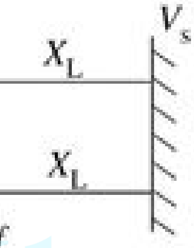
δ_{max}

$\underline{\underline{\omega = \omega_s}}$

$= 0.$	decel		not change.
$P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2}\sin\delta$ $(\delta > \delta_0)$ $= +ve$	accel^n	$\omega > \omega_0$	δ will slowly increase.
$P = P_{s_2} - P_{e_2}(\omega_0 = \omega)$ $= P_{s_2} - P_{m_2}\sin\delta_1$ $= 0.$	neither accel^n nor decel^n	$\omega = \omega_{max}.$ $[\omega_{max} > \omega_s]$	due to (ii) & (iii) assumption of equal area criteria rotor angle will further increase

$= P_{S2} - P_{e2}$ $P_{S_2} - P_{m_2} \sin\delta$ $(\delta > \delta_1)$ $= -ve$.	deceleration	$\omega <$ ω_{max} but $\omega > \omega_s$	even though the synch. generator is experiencing the deceleration, but the rotor angle will be further increasing because $\omega > \omega_s$
$= P_{S_2} - P_{e2}$ $= P_{S_2} - \frac{EX}{X_{eq}} \sin\delta_{max}$ $= -ve$	deceleration	$\omega = \omega_s$	Rotor angle at this moment starts decreasing.

$= P_{s_2} - P_{m_2} \sin\delta$ $\delta < \delta_{max}$ $= -ve.$	decel^n	$\omega < \omega_s$	slowly
$P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin\delta_1$ $= 0$	neither decel^n nor accel^n	$\omega = \omega_{min}$ $[\omega_{min} < \omega_s]$	Due to (ii) & (iii) assumptions of equal area criteria the rotor angle will further decrease
$P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin\delta$ $[\delta < \delta_1]$.	accel^n	$\omega > \omega_{min}$ but $\omega < \omega_s$	eventhough the synch. generator is having accel^n, but the actual speed is less than synch. speed so that rotor angle will be further decreasing.



$$P_s = Mechanical\ input$$

$$P_{e1} = Electrical\ Output\ before\ fault$$

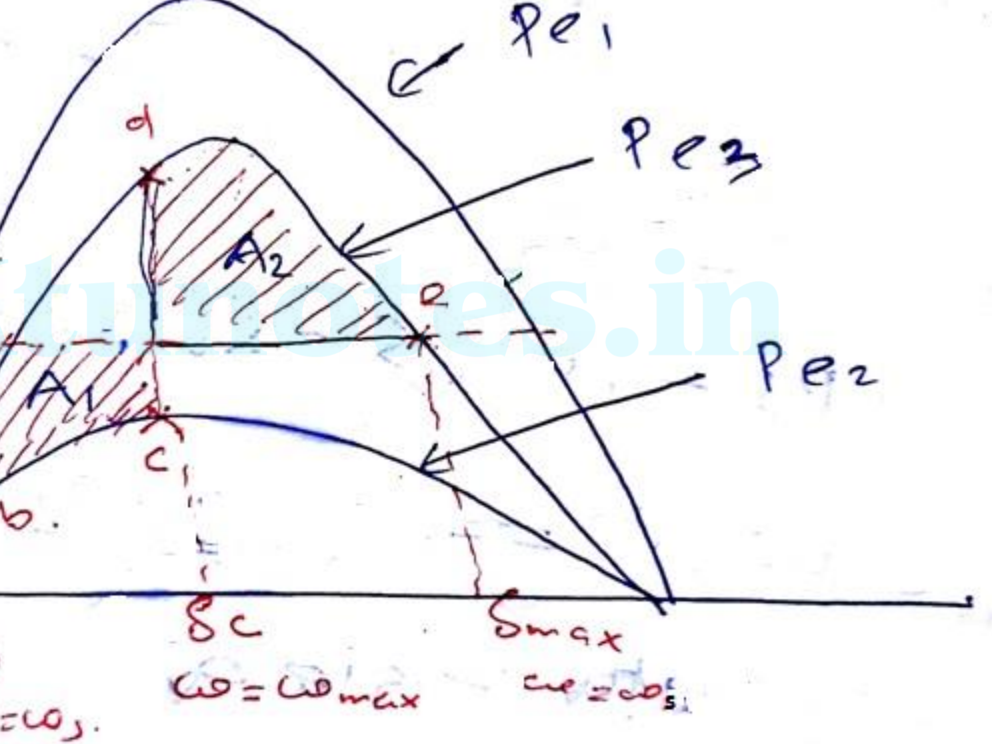
$$P_{e2} = Electrical\ Output\ during\ fault$$

$$P_{e3} = Electrical\ Output\ after\ fault$$

$$P_{e1} = \frac{EV}{X_{1eq}} Sin\delta_0 = P_{m1} Sin\delta_0$$

$$P_{e2} = \frac{EV}{X_{2eq}} Sin\delta_1 = P_{m2} Sin\delta_1$$

$$P_{e3} = \frac{EV}{X_{3eq}} Sin\delta_2 = P_{m3} Sin\delta_2$$



Pe1
Pe3
Pe2
d
A2
e
A1
c
b
δc
δmax
ω = ωmax

eration(Ps=Pe)

eleration nor Deceleration

hange

oint 'b' on another power angle
SSSL less than previous.

P_s

ve ,Accelerating power,Pacc

< P_s)

ses($\omega > \omega_s$)

>ω_s, δ increases $\rightarrow$ Pe further

point ‘d’ on another power angle
g SSSL more than previous, but
st.

ve ,Decelerating Power,Pdec

Pe > Ps)

ases(Still $\omega>\omega_s$)

ecrease from ω_{max}

$\omega>\omega_s$, δ increases→Pe further deceleration (Pe should be less than

cal point for stability

s such that Ps>Pe→Always unstable

ation and ω decreases from ωs

on

this curve

ed beyond this point , system will

$= 0$ · swinging point B

$a \rightarrow$ where $\omega = \omega_1$ & δ_0

$e \rightarrow$ where $\omega = \omega_1$, $\Rightarrow \delta_{max}$

δ_{max},

$$\delta + \int_{\delta_c} P_a \, d\delta = 0.$$

$$-P_{e}) \, d\delta + \int_{\delta_c}^{\delta_{max}} (P_s - P_{e}) \, d\delta = 0$$

$\delta_{max} \Rightarrow$ I/P $\rightarrow P_s$

O/P $\rightarrow P_{e3}$

$$\delta + P_{m2}\cos\delta\Big]_{\delta_0}^{\delta_c} + \Big(P_s\delta + P_{m3}\cos\delta\Big)_{\delta_c}^{\delta_{max}} = 0$$

$$\delta_c - P_s\delta_0 + P_{m2}\cos\delta_c - P_{m2}\cos\delta_0$$

$$P_s\delta_{max} - \cancel{P_s\delta_c} + P_{m3}\cos\delta_{max} - P_{m3}\cos\delta_c$$

$$(\delta_{max} - \delta_0) + P_{m3}\cos\delta_{max} - P_{m2}\cos\delta_0$$

$$= P_{m3}\cos\delta_c - P_{m2}\cos\delta_c$$

$$= \cos\delta_c (P_{m3} - P_{m2})$$

$$= \cos^{-1}\left[\frac{P_s(\overset{\text{rad}}{\delta_{max} - \delta_0}) + \overset{\text{degrees}}{P_{m3}\cos\delta_{max} - P_{m2}\cos\delta_0}}{P_{m3} - P_{m2}}\right]$$

decr. degrees.

$= P_{m1} \sin \delta_0$

$\sin^{-1}\left(\frac{P_s}{P_{m1}}\right)$ elect. degrees

$\delta_{max} = \sin^{-1}\left(\frac{P_s}{P_{m_3}}\right)$ elect. degrees.

...ng the critical clearing angle, the
...inging angle made by the synch.
...r. will be more than 90°'s.

$P_s = P_{e_3} = P_{m_3} \sin(180° - \delta_{max})$

$\frac{P_s}{P_{m_3}} = \sin(180 - \delta_{max})$

$180 - \delta_{max} = \sin^{-1}\left(\frac{P_s}{P_{m_3}}\right)$

$\delta_{max} = 180 - \sin^{-1}\left(\frac{P_s}{P_{m_3}}\right)$

$\delta_{max}(rad) = \delta_{max} \times \frac{\pi}{180}$

ver system analysis-
merical Problems

) Hz four-pole turbogenerator rated 20 MVA, 13.2 kV has an inertia
/kVA. Determine the K.E. stored in the rotor at synchronous speed.
n if the input less the rotational losses is 25000 HP and the electric
kW.

ince

$$H = \frac{\text{Stored energy in megajoules}}{\text{Machine rating in MVA}}$$

n the rotor in megajoules = $H \times$ Machine rating in MVA

$$= 9 \times 20 = 180 \text{ megajoules} \quad \textbf{Ans.}$$

ating power = Mechanical power input – Electrical power output

$$P_a = P_i - P_G$$

$$= 25000 \times 0.735 - 15000$$

$$= 18375 - 15000 = 3375 \text{ kW} = 3.375 \text{ MW}$$

ration is

$$\frac{d^2\delta}{dt^2} = \frac{P_a}{M} = \frac{3.375}{M}$$

$$M = \frac{GH}{\pi f} = \frac{20 \times 9}{\pi \times 50} = 1.146 \text{ megajoules-sec/radian}$$

$$\frac{d^2\delta}{dt^2} = \frac{3.375}{1.146} = 2.945 \text{ rad/sec}^2 \quad \textbf{Ans.}$$

chronous generator having an

age 1.2 pu, H = 5.2 MJ/MVA and a

f 0.4 pu is connected to an infinite

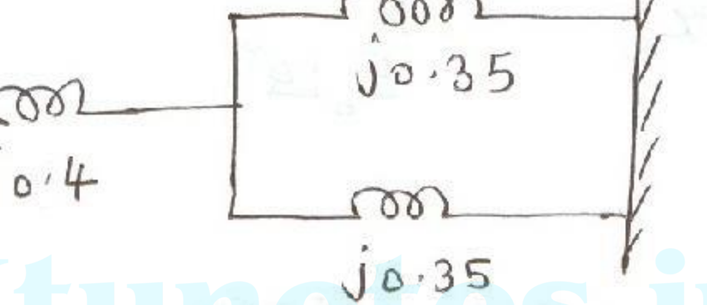
a double circuit line, each line of

35 pu. The generator is delivering

r and the infinite bus voltage is 1.0

ne: maximum power transfer,

operating angle



$$\frac{0.35}{2} = 0.575 \text{ pu}$$

$$E_b = \frac{1.2 \times 1.0}{0.575} = 2.087 \text{ pu}$$

δ_o

$$\frac{P_e}{P_{\max}} = \sin^{-1}\left(\frac{0.8}{2.087}\right) = 22.54^\circ.$$

Hz generator is delivering 50% of the power that it is capable of
mission line to an infinite bus. A fault occurs that increases the
erator and the infinite bus to 500% of the value before the fault.
the maximum power that can be delivered is 75% of the original
e the critical clearing angle for the condition described.

t the reactance is 500%

before the fault

ult is isolated, the maximum

can be delivered is 75% of the

ximum value.

$_{m1}$

aring angle

$$P_s(\delta_{max}-\delta_0)+P_{m3}\cos\delta_{max}-P_{m2}\cos\delta_0 \over P_{m3}-P_{m2}\Bigg]$$

P_{m1}

$$P_s/P_{m1}(\delta_{max}-\delta_0)+P_{m3}/P_{m1}\cos\delta_{max}-P_{m2}/P_{m1}\cos\delta_0 \over P_{m3}/P_{m1}-P_{m2}/P_{m1}\Bigg]$$

2

P_{m1}

$= 0.667$

$^{0} = 138.2^{0} = 2.412\,radian$

$$\left.(\delta_{\max} - \delta_0) + \frac{P_{m3}}{P_{m1}} \cos\delta_{\max} - \frac{P_{m2}}{P_{m1}} \cos\delta_0 \over \frac{P_{m3}}{P_{m1}} - \frac{P_{m2}}{P_{m1}}\right]$$

$$\left..412 - 0.5236) + 0.75\cos 138.2^{0} - 0.2 \times 0.866 \over 0.75 - 0.2\right]$$